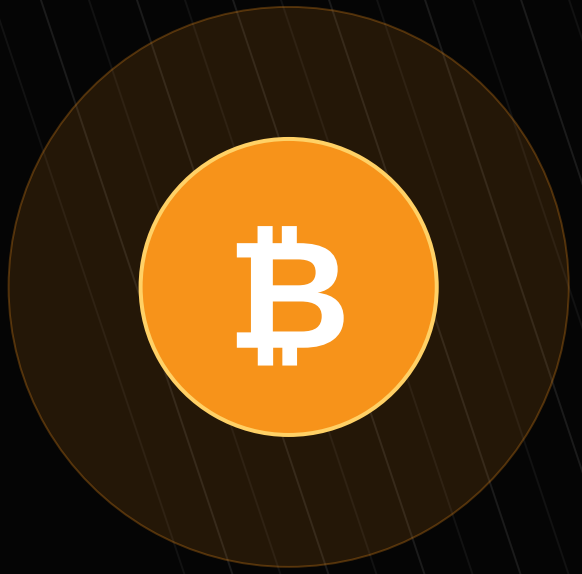
PROJECT WALKTHROUGH

BitScan

Bitcoin explorer architecture, data flow, and how each frontend detail is sourced from the Rust API and BitRPC.



Current state

No local indexer yet. The backend asks a Bitcoin RPC provider for live chain, block, transaction, mempool, and UTXO information.

Rust Axum API • Next.js frontend • BitRPC JSON-RPC

WHY THIS APP

BitScan

Bitcoin data, made traceable

BitScan turns raw Bitcoin Core RPC responses into readable screens while keeping every important value explainable from source code.

Explorer

Chain tip, latest blocks, block detail pages, transaction pages, and address lookups.

UI

Backend

Rust handlers normalize BitRPC/Bitcoin Core data into typed JSON models for the frontend.

API

Explainability

Fields such as block time, merkle root, nonce, confirmations, and fees map directly to specific RPC calls.

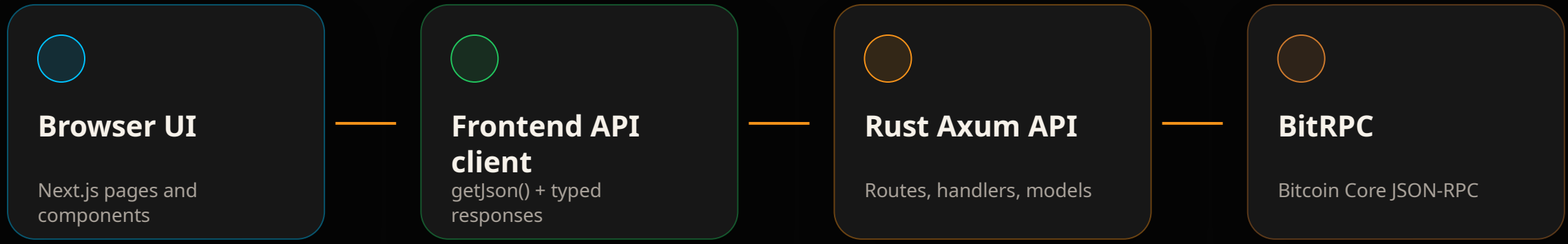
RPC

ARCHITECTURE

BitScan

One request path, four layers

The browser never talks to BitRPC directly. It asks the Next.js app, which calls the Rust API, which sends Bitcoin JSON-RPC methods to BitRPC.



Current architecture has no local database or indexer: results are resolved live through the RPC provider.

ROUTE MAP

BitScan

Frontend pages to Rust endpoints

The UI is composed from a small set of typed REST-style endpoints exposed by the Rust API.

Page	Rust endpoint	Used for
/	/api/tip, /api/mempool, /api/blocks	Dashboard, chain tip, mempool, latest blocks
/block/[id]	/api/block/{id}, /api/block/{id}/txs	Header fields, block stats, paged transaction summaries
/tx/[txid]	/api/tx/{txid}?block_hash=…	Transaction metadata, inputs, outputs, fee from decoded raw tx
/address/[address]	/api/address/{address}	Validation, spendable UTXO set, balance
/search	/api/search?q=…	Classifies height, hash, txid, or address

BLOCK DATA

BitScan

How the dashboard and block page are filled

Most block-facing UI data comes from getblockchaininfo, getmempoolinfo, getblockhash, and getblock.

Dashboard calls

- **Current chain tip**

  Rust /api/tip calls getblockchaininfo for blocks, bestblockhash, chain, and verificationprogress.

- **Mempool**

  /api/mempool calls getmempoolinfo for tx count, bytes, usage, and mempoolminfee.

- **Latest blocks**

  /api/blocks resolves heights using getblockhash, then fetches summaries with getblock(hash, 1).

Block detail fields

Time	block.time	Merkle root	block.merkleroot
Bits	block.bits	Nonce	block.nonce
Difficulty	block.difficulty	Weight	block.weight
Confirms	confirmations	Tx count	tx.length

TRANSACTION DATA

BitScan

Where transaction details come from

Transaction screens start with getrawtransaction. When available, a block_hash is passed to providers without txindex so old block transactions can still resolve.

Lookup flow

1. Validate txid is 64 hex characters
2. Call getrawtransaction(txid, true, block_hash?)
3. Map tx fields into TransactionDetail
4. Resolve previous outputs to estimate fees

Frontend cards

Confirmations	tx.confirmations	**Output total**	sum(outputs.value_sat)
Virtual size	tx.vsize	**Fee**	input_total - output_total
Inputs	tx.vin + previous_output	**Outputs**	tx.vout scriptPubKey

SEARCH + ADDRESS

BitScan

Classification before rendering

BitScan classifies the query first, then redirects the user to the right page or shows a handled error state.

Search decision tree

number	getblockhash(height)
64 hex	try getblock(hash), else getrawtransaction(txid, true)
address-like	validateaddress(address)

Address lookup

Current backend returns spendable UTXO-set data, not full historical address analytics.

- **validateaddress**

  Confirms the address belongs to the active Bitcoin network.

- **scantxoutset**

  Scans the node UTXO set for addr(address), returning unspents and balance.

CURRENT LIMITS

BitScan

What is live today, and what comes next

The BitRPC-only version is simple and deployable, but full explorer behavior needs either an indexer or a provider with txindex/address-history support.

Works now

Chain tip, mempool, block pages, block transaction pages, tx pages when block hash is known, and UTXO balances.

Watchouts

Provider limits, missing txindex, no full address history, and slow pages when many RPC calls are required.

Best next move

Add caching plus an indexer layer for addresses, transaction summaries, fee analytics, and faster search.

Pitch line: BitScan is a Bitcoin explorer that shows the data and teaches where that data comes from.